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Treatment use, perceived need, and barriers to seeking treatment for substance abuse and mental health problems among older adults compared to younger adults

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ABSTRACT

Background: This study examined age group differences in and correlates of treatment use and perceived treatment need for substance use disorders (SUD) and mental health (MH) problems as well as self-reported barriers to treatment among people 65+ years old vs. 26–34, 35–49, and 50–64 years old.
Methods: Data are from the 2008 to 2012 National Survey on Drug Use and Health (NSDUH) (N = 96,966). Age group differences were examined using descriptive bivariate analyses and binary logistic regression analyses.
Results: The 65+ age group was least likely to use treatment and perceive treatment need, but the 50–64 age group was more similar to the younger age groups than the 65+ age group. Controlling for age, other predisposing, and enabling factors, alcohol and illicit drug dependence and comorbid SUD and MH problems increased the odds of SUD treatment use. Of MH problems, anxiety disorder had the largest odds for MH treatment use. Bivariate analyses showed that lack of readiness to stop using and cost/limited insurance were the most frequent barriers to SUD and MH treatment, respectively, among older adults, and they were less likely than younger age groups to report stigma/confidentiality concerns for MH treatment.
Conclusions: Older adults will become a larger portion of the total U.S. population with SUD and/or MH problems. Healthcare providers should be alert to the need to help older adults with SUD and/or MH problems obtain treatment.

1. Introduction

Given longer life spans and the aging baby boom generation, which has had greater exposure to illicit drugs than previous generations, the number of Americans aged 50 years and older with a substance use disorder (SUD) has been increasing and is projected to double from 2.8 million (annual average) in 2002–2006 to 5.7 million in 2020 (Han et al., 2009). Increases are projected to occur in both genders and all racial/ethnic groups (Blow and Barry, 2012). With increasing SUDs among those aged 50+ years, help-seeking and treatment admissions in this age group have also steadily increased during the past decade (Arndt et al., 2011; Oleski et al., 2010; Sacco et al., 2013; Wu and Blazer, 2011). Those aged 50+ were 6.6% of all SUD treatment admissions in 1992 and 12.2% in 2008 (Substance Abuse and Mental Health Services Administration [SAMHSA], 2010). Alcohol remained the primary substance of abuse for most admissions. There was also a significant increase in admissions for a primary drug problem other than alcohol (e.g., 7.2% in 1992 to 16.0% in 2008 for primary heroin abuse).

The prevalence and treatment of mental health (MH) problems among older adults are significantly lower than for younger adults, but there has also been an overall upward trend in diagnosis and treatment among older adults. Medicare claims data from 1992 to 2005 showed that the proportion of older adults who received a depression diagnosis doubled from 3.2% to 6.3%, with rates increasing substantially across all demographic subgroups (Akincigil et al., 2011). Of those diagnosed with depression, the proportion receiving any treatment (pharmacotherapy and/or psychotherapy) steadily increased from 64.6% to 71.4% during the study period. Data from the 2004 to 2007 National Survey of Drug Use and Health also found that of the 4.7% of the population aged 65+ that had past-year serious psychological distress,

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37.7% received MH services −4.8% received inpatient services, 15.8% received outpatient services, and 32.1% received prescription medications (Han et al., 2011). Use of combined treatment modalities may also be increasing. A study of community-residing older adults aged 70+ years found that antidepressant medication combined with other services from a MH professional increased between 1998 and 2007, while antidepressant use alone decreased (Barry et al., 2012).

The growing evidence base on pharmacotherapy and psychotherapy for SUD and MH problems shows that older adults have the same or more positive treatment outcomes than their younger counterparts (Blazer, 2003; Goncalves and Byrne, 2012; Karlin et al., 2013; Kuerbis and Sacco, 2013; Satre et al., 2012; Weiss and Petry, 2013). A study using National Comorbidity Survey Replication data also found that more than 80% of individuals aged 55+ years reported positive attitudes toward seeking MH services, and more than 70% had positive beliefs about treatment efficacy; these rates were comparable to or higher than those of younger age groups (Mackenzie et al., 2008). However, most older adults do not access SUD and MH treatment (Huang et al., 2013; SAMHSA, 2010). Low treatment utilization rates among older adults may be attributable to denial of problems (thus lack of readiness for treatment), stigma/confidentiality issues, self-sufficiency beliefs (i.e., they could handle the problem themselves), lack of knowledge about treatment programs, and treatment cost (Conner et al., 2010; Jimenez et al., 2013; Keyes et al., 2010; Mackenzie et al., 2008, 2010; Oleski et al., 2010).

Older adults with untreated SUDs and/or mental disorders have substantially greater medical comorbidity than those without these disorders (Chapman and Perry, 2008; Lin et al., 2011). Given increasing rates of SUD and mental disorders among the growing population of older adults, it is important to examine correlates of their treatment use as well as self-reported reasons for not seeking treatment. Using multi-year epidemiologic data, this study compared adults aged 65+ years with younger adults aged 26–34, 35–49, and 50–64 years old on (1) rates of SUD and MH treatment use and perceived treatment need in the preceding 12 months; (2) sociodemographic, health, and clinical correlates of treatment use and perceived treatment need; and (3) self-reported reasons for not seeking treatment among those with perceived treatment need.

Andersen's (1995) behavioral model of health service use posits that health service access and utilization is a function of (1) predisposition to service use, including individuals' demographic characteristics and attitudes, values, and knowledge about health and health services that might impact their perceptions of need and health service use; (2) personal and system-level factors that enable or impede use, including social support, income, insurance coverage, availability and accessibility of services, transportation to treatment, and waiting times; and (3) individuals' perceived treatment need, which tends be more important than predisposing and enabling factors. For example, alcohol problem severity has been found to be a stronger factor for treatment seeking than sociodemographic characteristics (Saunders et al., 2006; Weisner et al., 2002). Based on the literature review and Andersen's behavioral model, the following hypotheses are proposed: (H1) controlling for other predisposing, enabling, and need factors, older adults (65+ years) will be less likely than younger adults to have used SUD and MH treatment and perceived treatment need; (H2) regardless of age, symptom severity (i.e., alcohol and drug dependence than abuse and MH diagnoses) will increase the odds of treatment use and perceived treatment need; and (H3) older adults will be more likely than younger adults to report lack of readiness, stigma, self-sufficiency beliefs, and lack of knowledge about treatment programs as barriers to treatment use.

2. Material and methods

2.1. Data and sample

Data came from the public use files of the 2008 to 2012 National Survey of Drug Use and Health (NSDUH), the largest population-based survey that measures substance use and MH problems and treatment use among the civilian, non-institutionalized, U.S. population aged 12 years or older (Inter-University Consortium for Political and Social Research [ICPSR], 2012). The survey included questions based on the fourth edition of the Diagnostic and Statistical Manual of Mental Disorders (DSM-IV) that were asked of respondents aged 18 years and older. Respondents were interviewed at their residence using computer-assisted self-interview and/or personal interview to increase the validity of reports of substance use and illegal activity (ICPSR, 2012).

To increase the study's power to detect low frequency events (e.g., SUD treatment), we combined five years of data. The survey sampling and data collection methods (including all the questions) for the variables examined in this study were the same for all five survey years. The data set groups respondents into six age brackets (12–17; 18–25; 26–34; 35–49; 50–64; and 65+). The total number of respondents aged 12+ was 283,049. This study focused on the 96,966 respondents aged 26+. NSDUH's multi-stage area probability sampling design made it unlikely that any duplication of survey respondents occurred in the pooled five years of survey data.

2.2. Measures

Past-year SUD: Included alcohol and illicit drug abuse or dependence that met DSM-IV diagnostic criteria. Alcohol and marijuana abuse or dependence were determined only if respondents reported using the substance on more than 5 days in the past year. Diagnostic criteria for other illicit drugs were applied regardless of days of use.

Past-year MH problems: Included major depressive episode (MDE) that met DSM-IV diagnostic criteria; anxiety disorder told by a doctor/other medical professional to the respondent; worst month serious psychological distress (SPD; yes = 1 vs. no = 0), measured with the six-item Kessler Psychological Distress Scale (K6; SPD = 1 if K6 $\geq$ 13; Kessler et al., 2003); and self-reported serious suicidal ideation (yes = 1 vs. no = 0). Cronbach's alpha for the K6 among the study sample was 0.91.

Past-year SUD treatment use: Referred to treatment during the past 12 months, including detoxification and any other treatment for medical problems associated with alcohol or drug use, not counting cigarettes. SUD treatment venues included hospitals; other inpatient/residential settings; emergency departments; prisons/jails; outpatient mental health centers/facilities; doctors'/private therapists' offices; partial day hospitals or day treatment programs; and other places such as informal sources, self-help groups such as Alcoholics Anonymous or Narcotics Anonymous (AA/NA), methadone clinics, halfway house/group homes, work places, social services, spiritual or religious advisors, and telephone hotlines.

Lifetime SUD treatment use: Referred to any treatment or counseling received in the respondent's lifetime.

Past-year MH treatment use: This was measured with questions about staying overnight or longer in a hospital or other facility to receive treatment or counseling, and/or receiving any outpatient treatment or counseling for "any problem with emotions, nerves, or mental health (not including treatment for alcohol or drug use)" during the past 12 months. Respondents were also asked if they received "treatment, counseling, or support" from the following sources: acupuncturists or acupressurists; chiropractors; herbalists; in-person support group or self-help group; internet support group or chat room; spiritual or religious advisors, such as a pastor, priest, rabbi; telephone hotlines; and massage therapists. In the present study, MH treatment use refers to any treatment received from inpatient or outpatient settings and/or these alternative sources.

Perceived treatment need: Perceived need for SUD treatment was measured with the question: "During the past 12 months, did you need treatment for alcohol or drug use?" Those who received any SUD treatment were also asked: "During the past 12 months, did you need additional treatment for alcohol or drug use?" Perceived MH treatment need was measured with the question, "During the past 12 months, was there any time you needed mental health treatment or counseling for yourself but didn't get it?" Response categories were: yes = 1 and no = 0.

Self-reported reasons for not obtaining treatment or counseling: Those who perceived SUD treatment need but did not get treatment/additional treatment or counseling and those who perceived MH treatment need but did not get it were asked to select one or more reasons including readiness, treatment beliefs, stigma/confidentiality concerns, lack of knowledge about treatment programs, cost, lack of openings in a program, lack of time, and lack of transportation.

Sociodemographic and health status variables: Included age group, gender, race/ethnicity, marital status, educational level, employment status, income, and self-ratings of health on a 5-point scale (1 = excellent to 5 = poor).

2.3. Analysis

All analyses were conducted with Stata/MP 13's svy function to account for NSDUH's multi-stage, stratified sampling design. Stata's subpop command was used for all subsample analyses to ensure that variance estimates incorporated the full sampling design. For the 5-year pooled data set used, adjusted person-level analysis weights were created by dividing the final person-level analysis weights by five (the number of years of combined data) following NSDUH guidelines for combining sampling weights across multiple years. All estimates presented are weighted except for sample sizes. First, descriptive statistics (χ^2 tests and Bonferroni-corrected ANOVA) are used to present sample characteristics by age group: (1) for the entire sample; (2) for those with past-year SUD (i.e., alcohol or illicit drug abuse or dependence); and (3) for those with MH problems (MDE, anxiety disorder, SPD, or serious suicidal ideation). Second, binary logistic regression analyses including all potential predisposing, enabling, and need factors are used to identify correlates of SUD and MH treatment use and perceived treatment need among those with SUD and MH problems, respectively. The predisposing factors included were age, gender, and race/ethnicity. The enabling factors were marital status (a measure of social support), education level, employment status, and income. Need factors were health status (self-rated health) and clinical characteristics including SUD and MH diagnoses and comorbid SUD and MH. Third, χ^2 tests are used to compare age groups on self-reported reasons for not obtaining treatment during the past 12 months despite perceived treatment need.

3. Results

3.1. Sample characteristics

Table 1 shows that of all respondents, 6.94% had a SUD and 15.25% had a MH problem. As expected, the 65+ age group had the

Table 1
Sample characteristics: age group differences in sociodemographics, health, and past-year substance use disorders (SUD) and mental health (MH) problems (N = 96,966).

Weighted (%)	All 100	26–34 Years 18.55	35–49 Years 31.99	50–64 Years 29.44	65+ Years 20.02
Sociodemographics and health					
Female	52.13	50.39	51.05	56.38	52.13
Race/ethnicity					
Non-Hispanic White	69.11	58.82	64.25	73.64	79.75
Non-Hispanic Black	11.10	12.73	12.08	10.82	8.46
Non-Hispanic Asian	4.61	6.06	5.44	3.80	3.11
Hispanic	13.21	20.26	16.31	9.64	6.98
All other	1.97	2.13	1.92	2.11	1.70
Married	60.91	48.19	65.33	66.54	57.34
College (BA/BS) graduate	31.41	34.65	32.98	31.96	25.11
Employed	64.61	78.40	79.93	68.79	21.20
Income > 200% FPT[a]	69.97	61.80	69.70	75.48	69.86
Self-rated health (M, SE)[b]	2.42 (0.01)	2.11 (0.01)a	2.29 (0.01)b	2.52 (0.02)c	2.75 (0.03)d
SUD					
Alcohol abuse	2.95	5.83	3.41	2.01	0.94
Alcohol dependence	2.96	5.34	3.61	2.28	0.72
Illicit drug abuse	0.46	1.05	0.56	0.28	0.03
Illicit drug dependence	1.20	2.99	1.41	0.61	0.10
Illicit drug abuse except marijuana	0.25	0.52	0.36	0.11	0.03
Illicit drug dependence except marijuana	0.80	1.77	0.98	0.50	0.06
Any substance use disorder	6.94	13.50	8.25	4.90	1.78
MH problems					
Major depressive episode	6.48	7.66	7.69	7.19	2.41
Anxiety disorder	5.40	6.43	6.11	5.59	3.03
Serious psychological distress	9.21	13.73	11.05	7.94	3.95
Serious suicidal ideation	3.26	4.01	4.04	3.03	1.66
Any of the above	15.25	19.60	17.42	14.67	8.58
Comorbid SUD/MH problems	2.46	5.23	3.07	1.51	0.33

Note: statistics refer to percentages, unless otherwise specified. The main effect of age group is significant in all comparisons at p < 0.001.
[a] Federal poverty threshold.
[b] $F(3,58) = 388.13$, p < 0.001; a > b > c > d (Bonferroni corrected).

lowest rates of SUD (1.78%), MH problems (8.58%), and comorbid SUD and MH problems (0.33%) of all age groups, while the 26–34 age group had the highest rates of SUD (13.50%), MH problems (19.60%), and comorbidity (5.23%).

3.2. Age group differences in past-year SUD: Sociodemographics, SUD treatment use, and perceived need for treatment

Table 2 shows that of all respondents with SUD, 5.14% and 20.77% were in the 65+ and 50–64 age groups, respectively. The main effect of age group was significant for all sociodemographic variables and self-rated health. Although alcohol abuse or dependence were the primary SUDs for all age groups, the 65+ age group had the largest proportions of alcohol abuse and dependence combined and the smallest proportions of illicit drug abuse or dependence. A significantly smaller proportion of the 65+ age group received SUD treatment during the preceding year and in their lifetime and perceived the need for treatment; however, the proportions of those with SUD who received MH treatment did not differ by age group. Further analysis also showed that 58.42% of all those who used SUD treatment also used MH treatment; there was also no difference by age group ($\chi^2(3) = 2.21$, $p < 0.809$).

3.3. Age group differences in past-year MH problems: Sociodemographics, MH treatment use, and perceived treatment need

Table 3 shows that of all respondents with any MH problem, 11.27% and 28.33% were in the 65+ and 50–64 age groups, respectively. The main effect of age group was significant for sociodemographic variables except gender and self-rated health. SPD was the most prevalent MH problem in all age groups. While MDE was more prevalent than anxiety disorder for the three younger age groups, anxiety disorder was more prevalent than MDE for the 65+ age group. The main effect of age group was significant for MDE, anxiety disorder, SPD, and comorbidity. The proportions of older adults with any MH problem that received professional or alternative MH treatments and/or SUD treatment were also smaller than for other age groups. The main effect of age group was significant for all types of treatment use and perceived treatment need. Further analyses showed that 1.61% of the older adults who received MH treatment also received SUD treatment, while the proportions were 7.99%, 7.72%, and 5.32% for the 26–34, 35–49, and 50–64 age groups, respectively ($\chi^2(3) = 60.71$, $p < 0.001$).

3.4. Correlates of past-year SUD treatment use and perceived need among those with SUD: Logistic regression results

Table 4 shows that compared to the 35–49 age group, the 65+ age group and the 26–34 age group had lower odds (0.39 and 0.66, respectively) of SUD treatment use, but the 50–64 age group did not differ from the 35 to 49 age group. Men were more likely than women to have received treatment. Non-Hispanic Blacks and Hispanics were less likely than non-Hispanic Whites to have received treatment. Being married, college educated, employed, and having higher income all decreased the likelihood of treatment use. Alcohol abuse decreased the odds of receiving treatment, while all other SUD diagnoses and comorbid SUD and a MH problem increased the odds.

With respect to perceived SUD treatment need, those who received treatment during the preceding year were less likely to perceive the need for more treatment even though they still met SUD criteria. Compared to the 35–49 age group, the 65+ and 26–34 age groups had lower odds (0.39 and 0.72, respectively) of perceived treatment need, but the 50–64 age group did not differ from the 35–49 age group. Men were more likely to perceive

Table 2
Those with any past-year substance use disorder (SUD): age group difference in sociodemographic/health characteristics, treatment patterns, and perceived need ($N = 8{,}331$).

Weighted (%)	All 100	26–34 Years 36.08	35–49 Years 38.02	50–64 Years 20.77	65+ Years 5.14	*p*
Sociodemographics and health						
Female	32.61	33.01	34.23	29.72	29.55	0.158
Race/ethnicity						<0.001
Non-Hispanic white	69.80	64.87	69.27	75.08	86.92	
Non-Hispanic black	11.94	12.14	11.98	13.27	4.77	
Non-Hispanic Asian	1.55	2.02	1.74	0.66	0.46	
Hispanic	14.17	18.80	14.58	7.48	5.73	
All other	2.54	2.17	2.43	3.51	2.13	
Married	44.19	28.99	51.10	51.48	70.33	<0.001
College (BA/BS) graduate	27.21	30.37	24.07	23.54	43.06	<0.001
Employed	71.33	79.37	74.47	62.13	28.93	<0.001
Income ≥200% FPT	64.89	63.15	63.76	66.05	80.73	
Self-rated health (M, SE)[a]	2.52 (0.01)	2.33 (0.02)a	2.54 (0.02)b	2.75 (0.04)c	2.77 (0.12)d	<0.001
SUD						
Alcohol abuse	42.50	43.16	41.30	41.03	52.64	0.042
Alcohol dependence	42.68	39.56	43.78	46.61	40.50	0.013
Illicit drug abuse	6.67	7.75	6.78	5.78	1.89	0.015
Illicit drug dependence	17.35	22.12	17.11	12.46	5.45	<0.001
Illicit drug abuse except marijuana	3.61	3.87	4.34	2.27	1.82	0.022
Illicit drug dependence except marijuana	11.53	13.09	11.88	10.19	3.53	0.002
Comorbid SUD/MH problems	35.47	38.71	37.25	30.75	18.62	<0.001
SUD Treatment (TX)						
Ever received in lifetime	27.91	24.71	30.94	30.43	17.81	<0.001
Received past year	10.45	9.47	12.27	10.53	3.56	<0.001
Perceived TX need	6.30	5.81	7.14	6.69	1.93	<0.001
Past-year MH treatment	33.55	30.60	35.17	36.15	31.73	0.072

Note: statistics refer to percentages, unless otherwise specified.
[a] Higher scores refer to worse self-rated health; $F(3,58) = 40.39$; a > b > c; a > d; b = d; c = d (Bonferroni corrected).

Table 3
Those with past-year mental health (MH) problems: age group differences in sociodemographic/health characteristics, treatment patterns, and perceived need (N = 16,974).

Weighted row (%)	All 100	26–34 Years 23.84	35–49 Years 36.56	50–64 Years 28.33	65+ Years 11.27	p
Sociodemographics and health						
Female	62.14	61.41	61.09	62.94	65.08	0.186
Race/ethnicity						<0.001
Non-Hispanic white	74.18	67.81	73.31	77.51	82.13	
Non-Hispanic black	9.82	11.81	10.33	8.40	7.47	
Non-Hispanic Asian	2.70	3.86	2.77	2.20	1.34	
Hispanic	10.71	14.03	11.26	8.53	7.41	
All other	2.59	2.50	2.33	3.36	1.66	
Married	46.92	38.44	54.55	54.14	46.92	<0.001
College (BA/BS) graduate	27.25	31.76	28.17	25.84	18.25	<0.001
Employed	57.77	71.62	67.16	51.62	13.49	<0.001
Income ≥ 200% FPT	59.95	56.78	60.23	61.63	61.79	0.012
Self-rated health (M, SE)[a]	2.81 (0.01)	2.46 (0.01)a	2.73 (0.03)b	3.08 (0.03)c	3.18 (0.04)d	
MH problem						
Major depressive episode	42.49	39.07	44.13	49.0	28.08	<0.001
Anxiety disorder	35.41	32.80	35.06	38.11	35.33	0.009
Serious psychological distress	60.4	70.06	63.44	54.10	45.97	<0.001
Serious suicidal ideation	21.39	20.46	23.18	20.66	19.39	0.090
Comorbid MH/SUD	16.15	26.67	17.63	10.26	3.87	<0.001
MH treatment (TX)						
No treatment received	41.86	47.07	40.12	35.65	52.11	<0.001
Professional MH care only	37.11	32.38	36.57	42.24	35.97	<0.001
Both professional and alternative MH care	15.03	13.59	16.46	16.77	9.08	<0.001
Alternative MH care only	5.99	6.96	6.85	5.33	2.84	<0.001
Perceived treatment need	21.16	27.53	24.29	16.84	8.36	<0.001
Past-year SUD treatment	4.93	5.89	6.18	4.15	0.94	0.072

Note: statistics refer to percentages, unless otherwise specified.
[a] Higher scores refer to worse self-rated health; $F(3,58) = 250.34$, $p<0.001$; a > b > c = d (Bonferroni corrected).

treatment need; however, there were no differences by race/ethnicity, education, employment status, income, and self-rated health. Alcohol and drug dependence and comorbidity increased the odds of perceived need, while neither alcohol nor drug abuse was a significant correlate, while (3.13, 3.65, and 2.83, respectively).

3.5. Correlates of past-year MH treatment use and perceived need among those with MH problems logistic regression results

Table 4 also shows that compared to the 35–49 age group, the 65+ age group and the 26–34 age group had lower odds of receiving MH treatment (0.56 and 0.75, respectively), while the 50–64 age group did not differ from the 35–49 age group. Men were less likely to have received treatment. Non-Hispanic Blacks, Hispanics, and non-Hispanic Asians were less likely than non-Hispanic Whites to have received treatment. College education and higher income increased treatment odds, while being employed decreased them. Of all MH problems, anxiety disorder resulted in the highest odds (10.75) of treatment use. Comorbidity was not a significant factor.

Those who received mental health treatment during the preceding year were more likely to perceive the need for treatment/more treatment. Compared to the 35–49 age group, the 65+ and 50–64 age groups had lower odds (0.41 and 0.62, respectively), but the 26–34 age group had higher odds (1.24) of perceiving treatment need. Men were less likely than women and Hispanics and non-Hispanic Asians were less likely than non-Hispanic Whites to perceive this need; but non-Hispanic Blacks did not differ from non-Hispanic Whites. Married people and those with higher incomes had lower odds of perceiving treatment need. MDE, SPD, and serious suicidal ideation, respectively, increased the odds of perceived need by two-fold, but anxiety disorder was not a significant correlate. Comorbidity also increased the odds (1.67).

3.6. Self-reported reasons for not getting SUD and MH treatment

Table 5 shows that age group effects were not significant in any reason for not seeking SUD treatment/more treatment, although some differences were identified among age groups. Those aged 65+ most frequently mentioned lack of readiness to stop using and lack of knowledge about treatment services and/or lack of openings in a program. For the 26–34, 35–49, and 50–64 age groups, treatment cost/limited insurance was reported most frequently, followed by lack of readiness to stop using, stigma/confidentiality concerns (including negative effect on job), and self-sufficiency beliefs.

Table 5 also shows that with respect to MH treatment, the 65+ age group was least likely to report stigma/confidentiality concerns (including fear of being committed or taking medications), treatment cost, not knowing where to go for treatment, lack of time, and lack of transportation/inconvenience, whereas the 26–34 age group was most likely to report stigma/confidentiality concerns and lack of knowledge about treatment venues. No age group differences were found for thinking treatment was not needed; self-sufficiency beliefs; and doubts about treatment effectiveness. Regardless of age group, treatment cost was the most frequently reported barrier to MH treatment among those with perceived need.

4. Discussion

The 65+ age group remained a small proportion of those with any SUD or MH problem. Study findings support H1, since controlling for other predisposing, enabling, and need factors, adults aged 65+ with SUD or MH problems were significantly less likely to have used treatment and to report perceived treatment need than those in the 35–49 age group. Also of interest was that the 50–64 year olds did not differ from 35 to 49 year olds in SUD treatment use, perceived SUD treatment need, and MH treatment use.

Table 4
Correlates of past-year SUD and MH treatment use and perceived treatment need among those with SUD or MH problems: odds ratios (OR).

	SUD (N = 8331)				MH (N = 16,974)			
	Treatment use vs. nonuse		Perceived treatment need vs. no need		Treatment use vs. nonuse		Perceived treatment need vs. no need	
	OR (SE)	95% CI	OR (SE)	95% CI	OR (SE)	95% CI	OR (SE)	95% CI
Past-year SUD treatment use			0.40 (0.09)***	0.26–0.63				
Past-year MH treatment use							1.95 (0.32)***	1.70–2.24
Age group								
26–34 years	0.66 (0.07)***	0.53–0.83	0.72 (0.09)*	0.55–0.93	0.75 (0.04)***	0.68–0.84	1.24 (0.08)**	1.10–1.40
50–64 Years old	0.86 (0.17)	0.59–1.27	0.98 (0.20)	0.65–1.49	1.14 (0.08)	0.99–1.31	0.62 (0.06)***	0.51–0.74
65+ Years old (35–49 years old)	0.39 (0.17)*	0.16–0.93	0.36 (0.12)**	0.18–0.69	0.56 (0.06)***	0.45–0.69	0.41 (0.07)***	0.29–0.57
Gender								
Male (female)	1.26 (0.14)*	1.00–1.58	1.54 (0.23)**	1.14–2.07	0.59 (0.04)***	0.52–0.66	0.73 (0.04)***	0.65–0.83
Race/ethnicity								
Non-Hispanic Black	0.72 (0.10)*	0.54–0.96	1.19 (0.21)	0.84–1.68	0.65 (0.07)***	0.53–0.80	1.02 (0.10)	0.84–1.24
Hispanic	0.60 (0.09)**	0.44–0.81	1.04 (0.20)	0.71–1.53	0.52 (0.06)***	0.42–0.65	0.73 (0.09)*	0.58–−0.94
Non-Hispanic Asian	0.54 (0.33)	0.16–1.85	1.08 (0.74)	0.28–4.24	0.42 (0.07)***	0.30–0.58	0.42 (0.13)**	0.22–0.77
Other (non-Hispanic White)	0.95 (0.24)	0.58–1.58	1.25 (0.54)	0.53–2.95	0.80 (0.11)	0.61–1.06	1.03 (0.19)	0.72–1.49
Marital status								
Married (unmarried)	0.62 (0.08)**	0.48–0.81	0.71 (0.09)*	0.55–0.92	0.96 (0.05)	0.87–1.07	0.85 (0.04)**	0.77–0.94
Education								
College graduate (no college degree)	0.56 (0.09)**	0.40–0.77	0.82 (0.15)	0.57–1.18	1.78 (0.12)***	1.55–2.03	1.08 (0.09)	0.92–1.27
Employment status								
Employed (not employed)	0.76 (0.08)*	0.61–0.95	0.87 (0.14)	0.63–1.18	0.75 (0.04)***	0.67–0.84	1.05 (0.07)	0.91–1.21
Income								
200+% Of FPT (<200% of FPT)	0.72 (0.08)**	0.57–0.90	0.91 (0.15)	0.65–1.27	1.27 (0.08)***	1.12–1.44	0.77 (0.05)***	0.68–0.88
Self-rated health	0.92 (0.06)	0.81–1.05	1.14 (0.10)	0.95–1.37	1.01 (0.02)	0.98–1.05	1.07 (0.07)	0.95–1.22
Alcohol abuse	0.64 (0.12)*	0.44–0.93	0.88 (0.20)	0.56–1.38				
Alcohol dependence	2.02 (0.30)***	1.50–2.72	3.13 (0.60)***	2.13–4.60				
Illicit drug abuse	1.47 (0.250)*	1.04–2.07	0.99 (0.34)	0.49–1.98				
Illicit drug dependence	2.64 (0.31)***	2.09–3.33	3.65 (0.57)***	2.67–4.97				
Major depressive disorder					2.28 (0.13)***	2.04–2.55	2.15 (0.11)***	1.94–2.39
Anxiety disorder					10.75 (0.77)***	9.32–12.41	1.11 (0.08)	0.96–1.28
Serious psychological distress					1.78 (0.10)***	1.59–2.0	2.17 (0.10)***	1.98–2.38
Serious suicidal ideation					1.46 (0.09)***	1.30–1.65	2.27 (0.15)***	1.98–2.60
Comorbid MH and SUD	2.28 (0.22)***	1.88–2.78	2.83(0.43)***	2.10–3.83	1.12 (0.07)	0.99–1.27	1.67(0.11)***	1.46–1.91
Constant	0.13 (0.03)***	0.08–0.22	0.01 (0.01)***	0.01–0.03	0.43 (0.04)***	0.35–0.54	0.07 (0.14)***	0.04–0.10

SUD treatment use: $F(18, 43) = 21.98$, $p < 0.001$; perceived SUD treatment need: $F(19, 42) = 23.47$, $p < 0.001$. MH treatment use: $F(18, 43) = 70.94$, $p < 0.001$; perceived MH treatment need: $F(19, 42) = 63.37$, $p < 0.001$.

* $p < 0.05$.
** $p < 0.01$.
*** $p < 0.001$.

H2 is also supported since controlling for predisposing and enabling factors, SUD and MH symptom severities were found to increase the odds of treatment use and perceived treatment need. Alcohol and drug dependence increased the odds of SUD treatment use and perceived treatment need, while alcohol abuse decreased the odds of SUD treatment use and neither alcohol nor illicit drug abuse was significantly associated with perceived treatment need. The significance of dependence diagnoses is expected since dependence is associated with greater substance-related problems (Dawson et al., 2010). A previous study found that even among individuals with alcohol dependence, the number of DSM-IV alcohol abuse symptoms endorsed was significantly associated with treatment seeking (Kuramoto et al., 2011). Our findings suggest that alcohol or drug abuse symptoms alone (i.e., without dependence symptoms) may not be sufficient to motivate people to seek treatment, and this may be especially true for older adults given the lower odds of their treatment use. Borok et al. (2013) also found that older (age 55+) at-risk drinkers who were primarily male, White, highly educated, and in good health did not reduce/change their drinking patterns if they did not think drinking was a problem, while most who reduced heavy drinking did so because they thought it would benefit them. More research is also needed to determine how motivation and readiness to change may differ by specific diagnostic criteria and by age group.

Study findings provide partial support for H3. For older adults with SUD, the lack of readiness to stop using was indeed the most frequently reported barrier to treatment use. Older adults largely have alcohol abuse or dependence diagnoses, and they may not perceive drinking as problematic or stigmatizing. For MH treatment, older adults most frequently reported treatment cost as a barrier, although they were less likely than younger adults to report cost and other any system-level barriers, probably because they have Medicare coverage. Kuramoto et al. (2011) found that public insurance coverage (Medicare, Tricare, Champus, VA, and/or military health) increased the odds of treatment seeking controlling for sociodemographics and symptom severity. For both SUD and MH problems, stigma does not appear to be a significant treatment barrier for older adults, while younger persons may be more concerned about stigma/confidentiality issues perhaps because they are more likely to worry about negative job impacts. A systematic review of the impact of mental health-related stigma on help-seeking found that ethnic minorities, youth, men and those in the military and health professions were disproportionately deterred by stigma (Clement et al., 2014). Although a lack of perceived treatment need

Table 5
Age Group difference in self-reported reasons for not getting treatment past-year among those with perceived need.

Weighted row (%)	SUD (*N* = 524)					
	All 100	26–34 Years 33.29	35–49 Years 43.09	50–64 Years 22.05	65+ Years 1.58	*p*
Stigma/confidentiality	20.57	25.41	20.0	14.99	12.39	0.589
Not ready to stop use	38.33	40.69	35.46	37.45	79.03	0.334
Did not think TX was needed at that time	5.10	4.28	3.45	11.79	0	0.082
Thought could handle by self	5.88	5.69	5.23	6.95	12.39	0.882
Did not think TX would help	4.39	2.88	5.52	4.81	0	0.737
Any of the above	56.85	61.1	54.07	54.28	79.03	0.516
Cost/limited insurance	43.92	48.75	42.74	40.37	24.07	0.589
Did not know where to go, no program, or no opening	16.60	16.56	12.97	23.0	27.21	0.361
Did not have time	5.12	4.32	5.53	5.91	0	0.872
No transportation/too far/inconvenience	7.65	3.62	10.62	7.20	17.83	0.216
Any of the above	56.4	58.76	58.58	49.60	41.09	0.583
Weighted row (%)	MH (*N* = 4039)					
	All 100	26–34 Years 31.02	35–49 Years 41.97	50–64 Years 22.55	65+ Years 4.45	*p*
Stigma/confidentiality	25.0	29.96	24.33	21.31	15.42	0.030
Did not think TX was needed at that time	5.19	6.03	4.31	4.43	6.43	0.567
Thought could handle by self	20.04	20.88	20.38	18.18	20.50	0.742
Did not think TX would help	7.91	7.77	7.39	8.65	10.11	0.762
Any of the above	44.21	47.87	43.15	40.95	45.13	0.277
Cost/no or limited insurance	57.59	58.40	58.85	58.76	34.07	<0.001
Did not know where to go	15.08	20.53	12.80	12.97	9.23	<0.001
Did not have time	12.86	14.97	13.87	9.23	7.11	0.041
No transportation/too far/inconvenience	3.72	3.16	3.72	5.11	0.61	0.040
Any of the above	72.58	74.96	74.27	70.84	48.98	<0.001

Note: statistics refer to percentages.

in itself may stem from stigma, self-sufficiency beliefs, and denial or lack of recognition of problems, NSDUH does not provide data on why people fail to perceive treatment need.

By examining both SUD and MH treatments, this study identified factors, in addition to age, that may be common or different for seeking these treatments. Racial/ethnic minorities, married individuals, and employed persons were less likely to use both SUD and MH treatment, but differences were found in the effects of gender, education, and income. Men and those with lower education and income were most likely to get SUD treatment, perhaps reflecting more mandated treatment for these problems (e.g., following DUI or drug-related arrests). Women and those with higher education and income were more likely to get MH treatment, perhaps reflecting higher social and individual acceptance of MH than SUD problems.

The study's limitations include the following: First, even with multi-year data, the number of older adults who perceived SUD treatment need is too small to allow broad generalizability of findings regarding self-reported treatment barriers. Second, the small number of older adults with perceived need also did not allow for tests of interaction effects between age group and other correlates of SUD treatment use and perceived need.

Despite these limitations, the findings provide the following practice and policy implications: First, since older adults were less likely than younger adults to report cost and stigma/confidentiality concerns as treatment barriers, healthcare providers might focus on other areas such as helping older adults understand treatment's health benefits and the serious deleterious effects that can occur from substance abuse due to metabolic and other changes that occur in older age. Older adults may not be ready to stop drinking because they may not understand its negative consequences on their health. Second, since the lower levels of SUD symptoms that characterize abuse do not appear to motivate people, older adults in particular, to seek help, more effort is needed to increase the public's knowledge and change attitudes and beliefs so that these lower-level symptoms are also recognized as negatively effecting health, mental health, and social functioning. Third, no-cost resources (e.g., on-line support/self-management programs; telephone hotline services; AA/NA meetings; prescription medication assistance programs) may be suggested for those with affordability issues. Fourth, universal public health campaigns targeting older adults are needed, as many may not recognize the need for treatment (e.g., thinking that depression is part of aging process) even though they may suffer from SUD and MH problems. Fifth, although SUD and MH problems tend to decrease with advancing age, the high prevalence of and treatment seeking for SUD and MH problems among 50–64 year olds means that treatment programs serving older adults need to be expanded in the near future as this group is fast becoming a larger portion of the total U.S. population with SUD and/or MH problems.

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Contributors

All three authors contributed to and have approved the final manuscript. Namkee Choi and Diana DiNitto conceptualized the paper and conducted the literature review. Namkee Choi conducted statistical analysis, and both Namkee Choi and Diana DiNitto wrote the manuscript. C. Nathan Marti reviewed the statistical analysis and the manuscript and commented on them.

Conflict of interest statement

No conflict declared.

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